DEVELOPMENT OF REAL TIME UNDERGROUND MONITORING SYSTEM USING UNMANNED VEHICLE

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ABSTRACT

The cave environment plays a crucial role in research across geology, biology, ecology, hydrology, and cultural anthropology. This project aims to develop an unmanned vehicle using IoT and Raspberry Pi technologies to monitor cave conditions and transmit data via GPS. The vehicle will detect environmental parameters such as air quality, temperature, humidity, lighting, objects, and soil Condition using integrated sensors and AI algorithms. Designed for safe navigation, the unmanned vehicle identifies hazards like air pollution, darkness, or water flow and autonomously retreats to avoid damage. The unmanned vehicles have become more reliable and practical for cave exploration. This system will assist in identifying dangers like collapses, flooding, landslides, and low oxygen zones, enhancing safety in cave exploration.

INTRODUCTION

- Cave environments are crucial for various fields like geology, biology, and anthropology, but their exploration is risky due to challenges like poor visibility, unstable structures, and low air quality. This study focuses on designing an unmanned vehicle for autonomous cave exploration using IoT and Raspberry Pi-based technologies.
- The vehicle, equipped with sensors and AI algorithms, can monitor critical parameters such as air quality, temperature, humidity, lighting and object proximity. It will transmit real-time data via GPS for remote monitoring and autonomously navigate hazards like pollution or flooding. By improving safety and data collection, this vehicle enhances cave exploration, mapping dangers like collapses and low oxygen zones efficiently.

OBJECTIVE(S) AND SCOPES

Objectives:

- The primary objective of this project is to develop an unmanned vehicle using Fiber Optic Technology (FOT) and Raspberry Pi systems for real-time monitoring of cave environments.
- The vehicle will measure critical environmental parameters such as air quality, lightning, obstacles, temperature, humidity, soil erosion, and toxic gases.
- Using AI algorithms for data processing and GPS for location tracking, the vehicle will provide accurate, real-time data to ensure safety in cave exploration and contribute to scientific research.

OBJECTIVE(S) AND SCOPES

Scope:

- This project covers the design and implementation of a comprehensive cave monitoring system using an unmanned vehicle.
- The system integrates advanced sensors, AI, and GPS technologies to measure environmental factors and transmit data to a ground station.
- The data will be displayed on a dedicated website through real-time graphs and images, with a safety status ("Safe" or "Unsafe") for cave conditions.
- The vehicle will enhance exploration safety, support diverse research disciplines, and adapt to unpredictable environmental conditions efficiently.

LITERATURE SURVEY-1

Title of the Paper	Author	Year of Publication	Publisher	Methods	Advantages	Limitations
Autonomous Robot Navigation in Underground Mines Using Machine Learning	A. W. Patterson, K. Ma	2023	Elsevier	Deep learning-based algorithms for autonomous navigation	Improved obstacle avoidance and path planning in dynamic environments	High computational power required, unknown terrains
Development of a Multi-Sensor Monitoring System for Underground Coal Mines	L. Jin, X. Li	2023	International Journal of Mining Science	Multi-sensor data fusion system (thermal, gas, vibration)	Accurate monitoring of hazardous conditions	Susceptibility to sensor failure, high maintenance costs

LITERATURE SURVEY-2

Title of the Paper	Author	Year of Publication	Publisher	Methods	Advantages	Limitations
Unmanned Aerial Vehicle-Based Underground Surveying for Geological Data Acquisition	Y. Wang, R. Liu	2022	IEEE	MobileNetV2 and SqueezeNet are used to develop a mobile applications.	High-resolution data capture, reduced risk for personnel	Limited operation time, GPS signal loss in deep tunnels .
Remote Monitoring System for Underground Coal Mine Safety	F. Li, Z. Huang	2021	IEEE	IoT-based real-time safety monitoring using wireless communication.	Timely data collection and hazard detection.	Network connectivity issues in deep mining environments

LITERATURE SURVEY-3

Title of the Paper	Author	Year of Publication	Publisher	Methods	Advantages	Limitations
Real-Time Monitoring and Control System for Underground Mining	H. Elmiligi, M. S. Hossain, N. S. O'Leary	2021	IEEE	Wireless sensor networks, embedded systems for real-time data transmission.	Enhanced real-time monitoring and decision-making capabilities.	Limited range and accuracy in complex underground structures
A Wireless Sensor Network for Underground Mine Environment Monitoring	S. Zhao, Q. Shi, Y. Xue	2020	Springer	Low-power wireless sensor network (WSN) for underground condition monitoring	Energy-efficient, scalable, suitable for harsh environments	High initial cost, challenges in data reliability in deep locations

LITERATURE SURVEY-4

Title of the Paper	Author	Year of Publication	Publisher	Methods	Advantages	Limitations
Real-Time Wireless Communication and Surveillance System for Underground Mines	J. Singh, M. Kumar	2021	Elsevier	ZigBee-based wireless communication with real-time video surveillance	Enhanced safety and operational monitoring	Limited bandwidth, signal interference in deep underground.
Underground Wireless Communication and Navigation for Autonomous Vehicles	D. Zhang, Y. Zhang	2019	IEEE	Wireless communication protocols for vehicle navigation	Real-time navigation in unstructured environments	High energy consumption, limited range in complex areas.

LITERATURE SURVEY-5

Title of the Paper	Author	Year of Publication	Publisher	Methods	Advantages	Limitations
Real-Time Air Quality Monitoring in Underground Mines Using Unmanned Ground Vehicles	R. Das, A. K. Dutta	2020	Elsevier	Integration of gas sensors with UGVs for air quality monitoring.	Real-time hazard detection and prevention.	Limited sensor accuracy, high operational cost.
Design of an Autonomous Monitoring System for Underground Mines Using UAV and UGV	B. H. Cho, J. Lee	2021	Springer	Coordination between unmanned aerial and ground vehicles for comprehensive monitoring	Enhanced data coverage, reduced risks to human operators	High system complexity, difficulty in maintaining communicatio n links

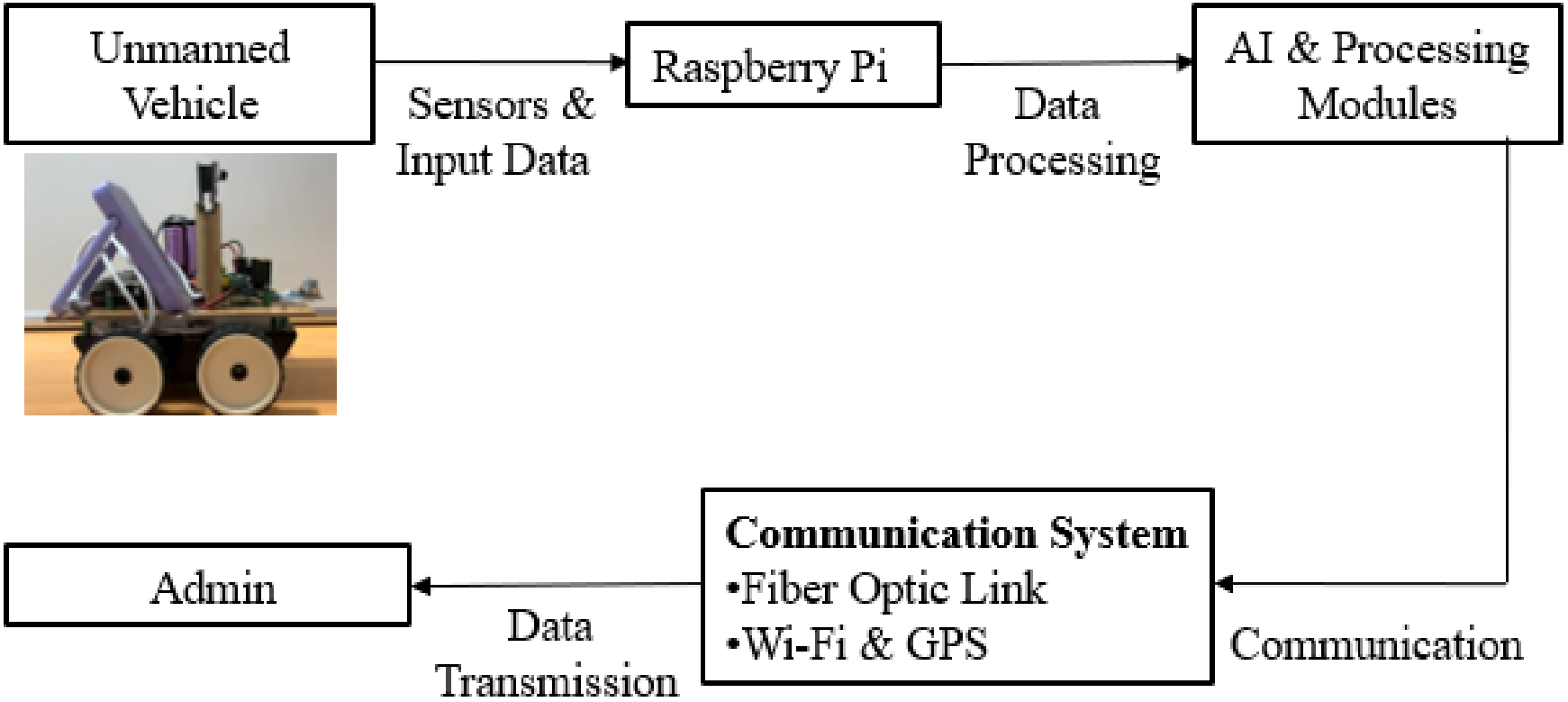
SUMMARY OF LITERATURE REVIEW

- The literature on underground monitoring systems using unmanned vehicles highlights significant advancements in wireless sensor networks (WSNs), IoT, and autonomous navigation technologies.
- Several studies focus on environmental monitoring, detecting parameters like air quality, water levels, and gas concentration. Machine learning-based algorithms have been employed for obstacle detection and navigation, improving safety in hazardous environments.
- Few solutions offer a comprehensive system that integrates monitoring, real-time hazard detection, and autonomous navigation.
- The proposed system aims to address these gaps by combining IoT, AI, and sensor technologies for efficient cave exploration and safety enhancement.

PROPOSED WORK

- The proposed system aims to develop an unmanned vehicle using IoT and Raspberry Pi technology to monitor cave environments with data transmitted via GPS.
- Equipped with various sensors and AI algorithms, the vehicle will track parameters such as air quality, temperature, humidity, lighting, and Soil Condition, autonomously navigating and avoiding hazards like polluted air or sudden water flow.
- This system will enhance cave exploration safety and contribute to research in multiple scientific fields by providing insights into potential dangers like collapses or low oxygen levels, leveraging advancements in materials, sensors, and technology for reliable, continuous data collection.

ARCHITECTURE DIAGRAM



NOVELTY IN PROPOSED WORK

- The novelty of the proposed system is its integrated approach combining IoT, Raspberry Pi technology, and advanced sensors with AI algorithms to monitor and navigate cave environments autonomously.
- Unlike existing solutions, this system provides comprehensive, real-time data on multiple environmental parameters and autonomously avoids hazards.
- Its use of GPS for continuous data transmission and real-time decision-making capabilities represents a significant advancement, enhancing safety and efficiency in cave exploration while contributing valuable insights to various scientific fields.

MODULES

- Environmental Sensing and Data Collection
- Autonomous Navigation and Obstacle Avoidance
- GPS Tracking
- User Interface and Data Visualization

MODULES DESCRIPTION

Environmental Sensing and Data Collection

This module involves using sensors to measure air quality, temperature, lighting, and object detection. The collected data is processed using AI algorithms to identify potential hazards and ensure safe exploration.

Autonomous Navigation and Obstacle Avoidance

The vehicle autonomously moves through the cave using depth sensor, and AI-driven path planning. It detects and avoids obstacles like rocks, water, and narrow passages while ensuring safe movement and retreat in hazardous conditions.

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GPS Tracking

A GPS module continuously tracks the vehicle's location inside or near the cave, providing real-time positioning data. This helps in navigation, retrieval, and data logging for research and safety monitoring.

User Interface and Data Visualization

A user-friendly dashboard (web or mobile) displays real-time sensor data, vehicle status, and hazard alerts. Researchers can monitor the cave environment remotely, analyze past data, and control the vehicle when necessary.

ALGORITHM TECHNIQUES

- This project integrates AI, IoT, and image processing for efficient cave exploration. The IoT framework collects real-time data on air quality, temperature, humidity, lighting, and water levels using Raspberry Pi and sensors.

- YOLO (You Only Look Once) algorithm enables real-time object detection for safe navigation.

- A dehaze algorithm enhances visibility in low-light and foggy conditions, improving obstacle detection. AI-based decision-making allows the vehicle to detect hazards like air pollution and flooding.

- An autonomous retreat mechanism ensures the vehicle avoids dangers. GPS tracking provides precise location data for navigation and research.

SYSTEM REQUIREMENTS

Hardware Requirements

- Microcontroller - 8051
- Ultrasonic Sensor -HC-SR04
- Air quality Sensor - MQ2 , MQ8 , MQ135
- Temperature and Humidity Sensor - DHT11
- Camera - ESP32 Camera
- Soil Condition Sensor - Capacitance Sensor

Software Requirements

- Processor - 64GB RAM
- GPS Module

Hardware Implementation:

```
#include "esp_camera.h"
#include "WiFi.h"
#include "app_httpd.h"
#include "camera_pins.h"
const char *ssid = "your_SSID";
const char *password = "your_PASSWORD";

void setup() {
    Serial.begin(115200);
    Serial.println("\nStarting ESP32-CAM...");

camera_config_t config;
    config.ledc_channel = LEDC_CHANNEL_0;
    config.ledc_timer = LEDC_TIMER_0;
    config.pin_d0 = Y2_GPIO_NUM;
    config.pin_d1 = Y3_GPIO_NUM;
```

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```
config.pin_d2 = Y4_GPIO_NUM;
    config.pin_d3 = Y5_GPIO_NUM;
    config.pin_d4 = Y6_GPIO_NUM;
    config.pin_d5 = Y7_GPIO_NUM;
    config.pin_d6 = Y8_GPIO_NUM;
    config.pin_d7 = Y9_GPIO_NUM;
    config.pin_xclk = XCLK_GPIO_NUM;
    config.pin_pclk = PCLK_GPIO_NUM;
    config.pin_vsync = VSYNC_GPIO_NUM;
    config.pin_href = HREF_GPIO_NUM;
    config.pin_sccb_sda = SIOD_GPIO_NUM;
    config.pin_sccb_scl = SIOC_GPIO_NUM;
    config.pin_pwdn = PWDN_GPIO_NUM;
    config.pin_reset = RESET_GPIO_NUM;
    config.xclk_freq_hz = 20000000;
    config.pixel_format = PIXFORMAT_JPEG;
    config.frame_size = FRAMESIZE_UXGA;
    config.fb_count = 1;
if (psramFound()) {
        config.jpeg_quality = 10;
```

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```
config.frame_size = FRAMESIZE_SVGA;
        config.fb_location = CAMERA_FB_IN_DRAM;
    }

if (esp_camera_init(&config) != ESP_OK) {
        Serial.println("Camera init failed!");
        return;
    }

WiFi.begin(ssid, password);
    Serial.print("Connecting to WiFi");
}
    Serial.println("\nWiFi Connected!");
    Serial.print("Visit: http://");
    Serial.println(WiFi.localIP());
startCameraServer();
}

void loop() {
    delay(10000);
```

OUTPUT

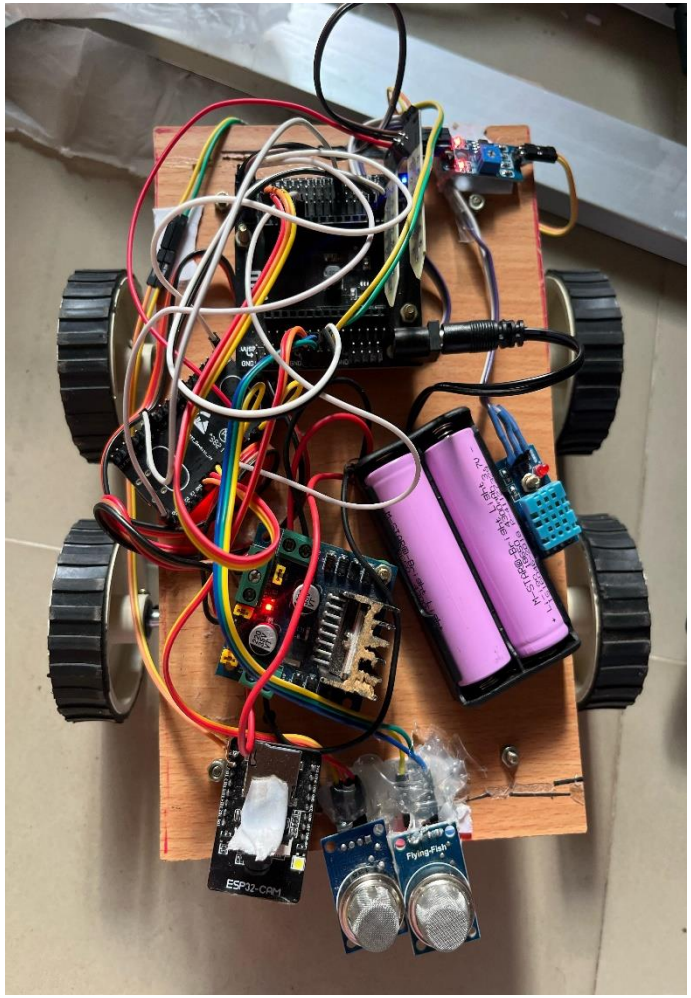

Fig 1. Vehicle Prototype

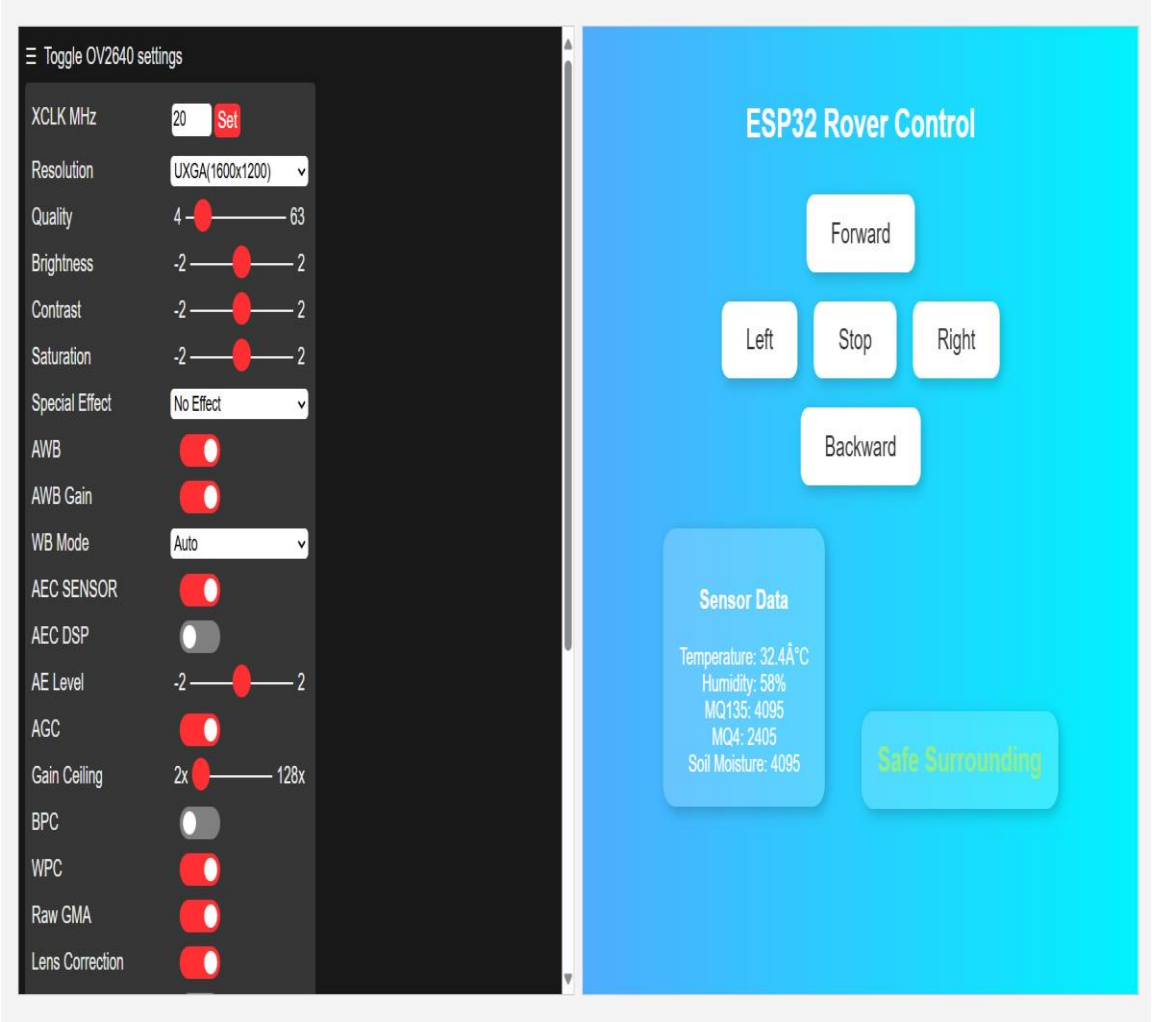

Fig 2. Control Website

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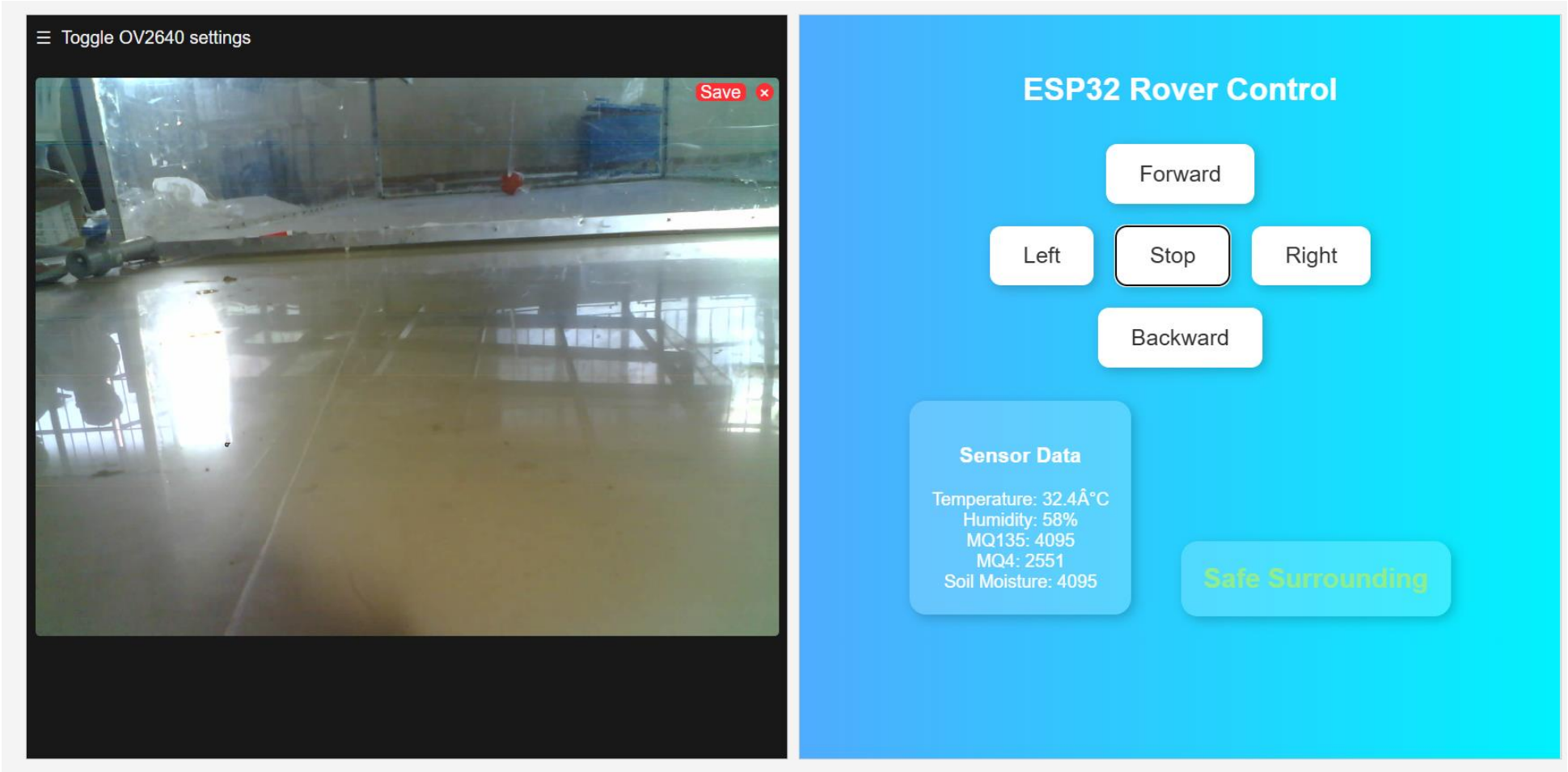

Fig 3. Video Monitoring

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Fig 4. Original Image Into Enhanced Image

Fig 5. Enhanced Image Into Obstacles Detection

CONCLUSION

- In conclusion, image processing offers immense potential in harsh environments, with strengths including robustness, adaptability, and automation. However, challenges such as sensitivity to conditions, hardware dependencies, and limited adaptability pose significant threats.
- Nonetheless, opportunities for technological advancements, emerging markets, integration with IoT and AI, environmental monitoring, and cross-industry collaboration indicate a promising future.
- We utilized an unmanned vehicle equipped with multiple sensors and devices. The vehicle involves employing the camera to capture images in a cave environment. Subsequently, in image processing, we must enhance the clarity of the images and reduce the amount of fog present.

FUTURE ENHANCEMENTS

In the future, the autonomous cave exploration vehicle could be upgraded with an efficient solar power system to enable continuous operation in remote, off-grid locations. This enhancement would minimize reliance on conventional battery charging, significantly extending the vehicle's operational lifespan during extended missions. The system would operate in two phases: solar energy integration and battery backup When solar energy is insufficient, the system would seamlessly switch to battery power, ensuring uninterrupted functionality in dark or low-light conditions.

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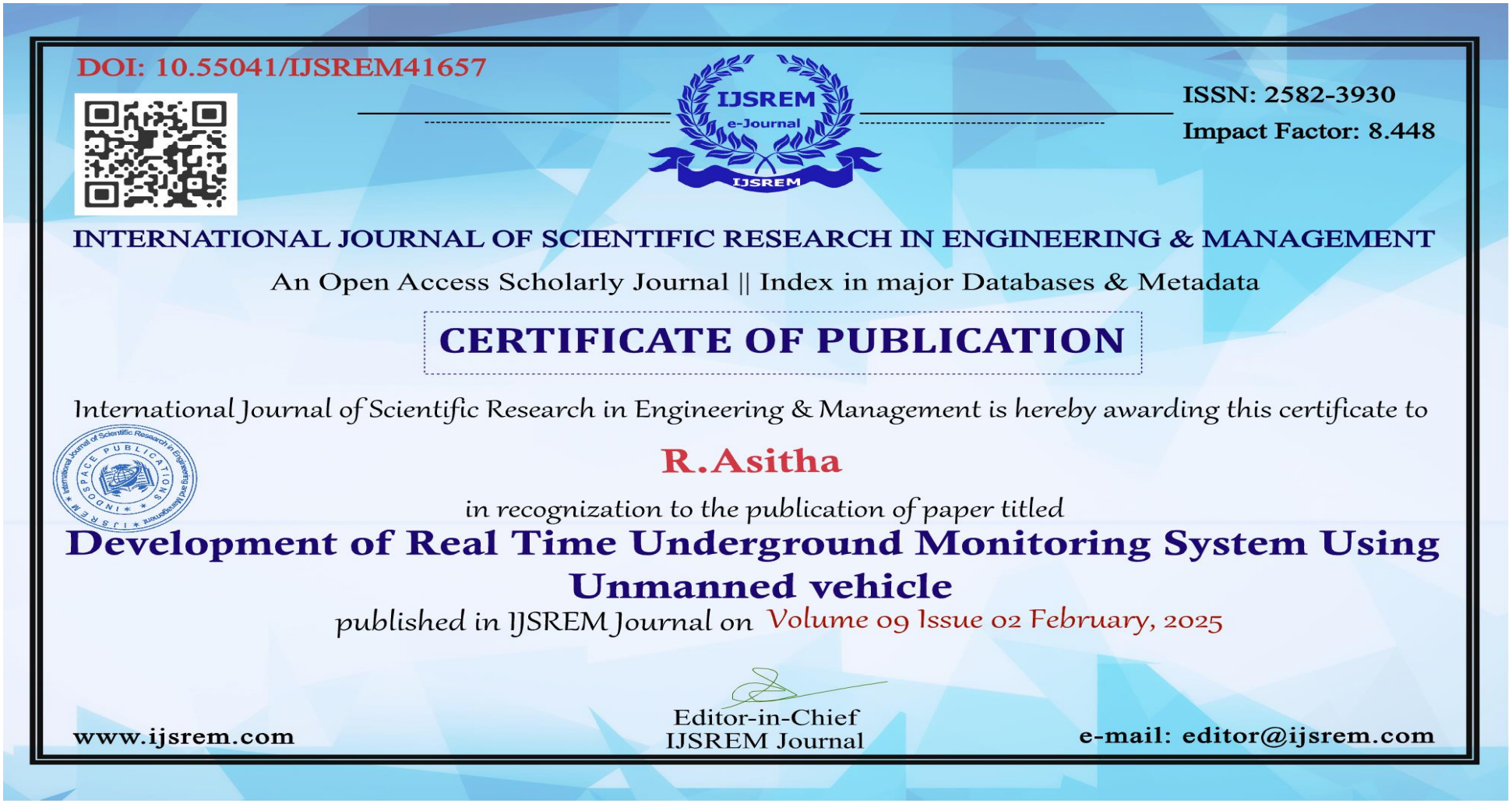

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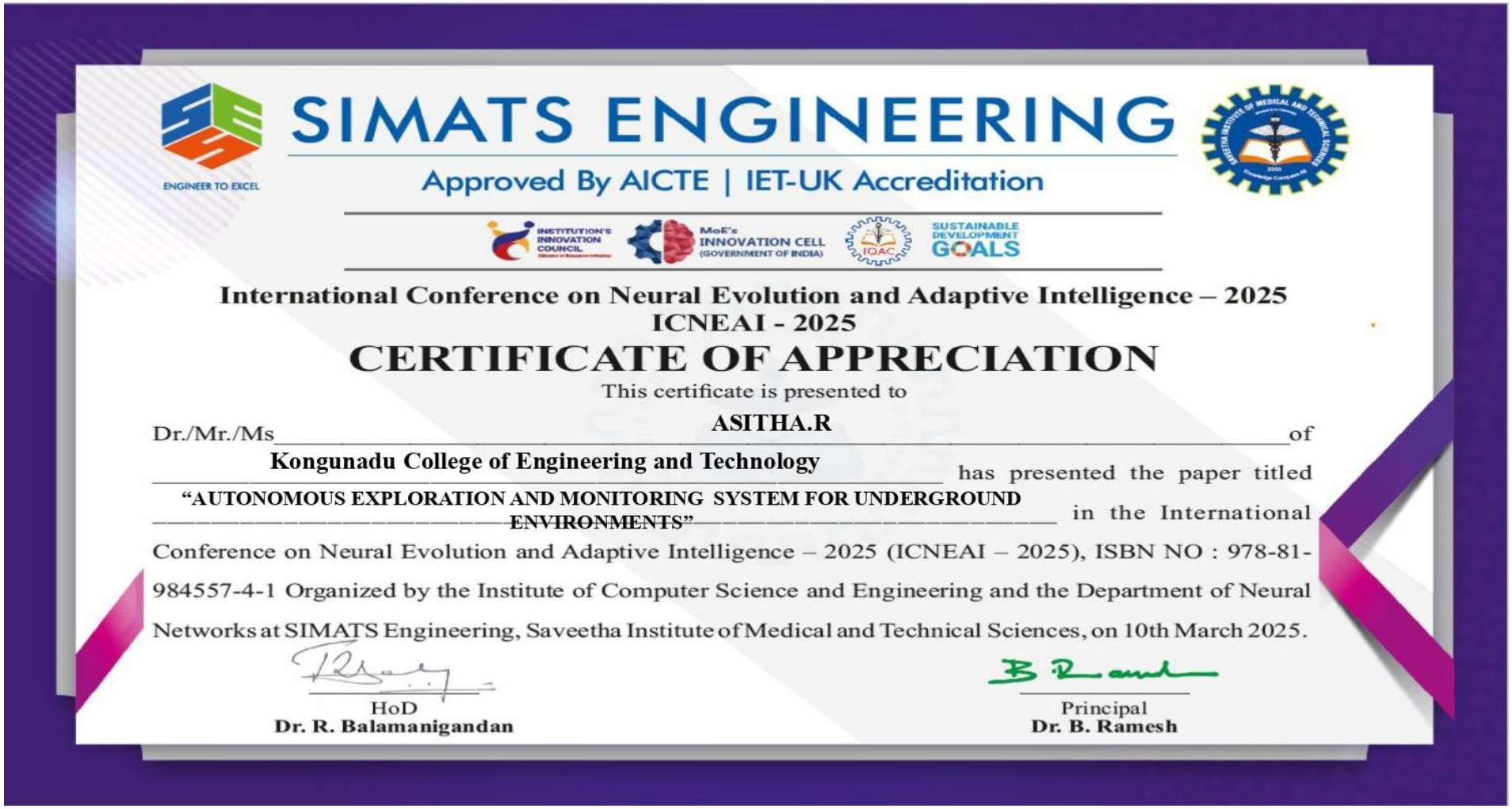

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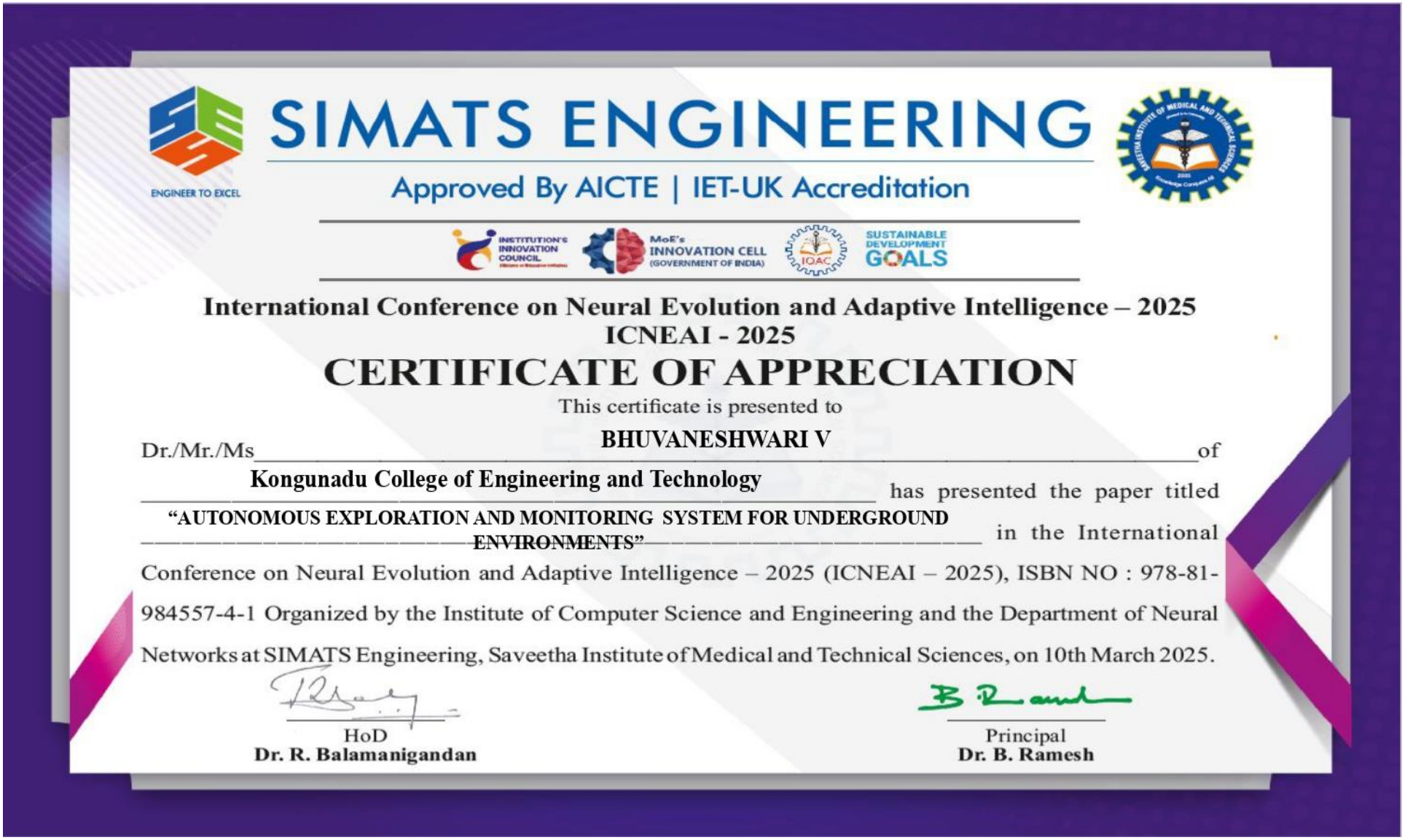

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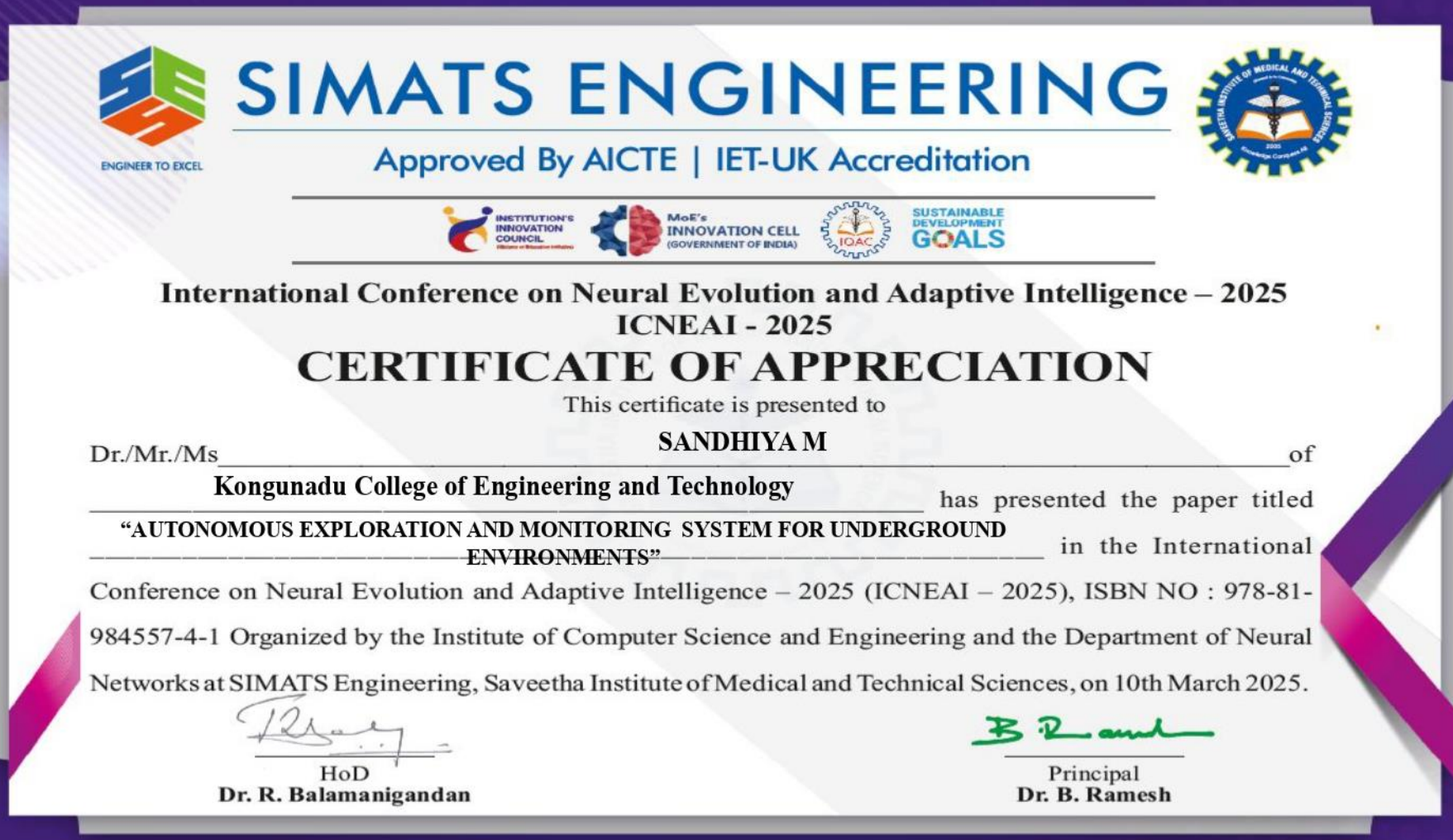

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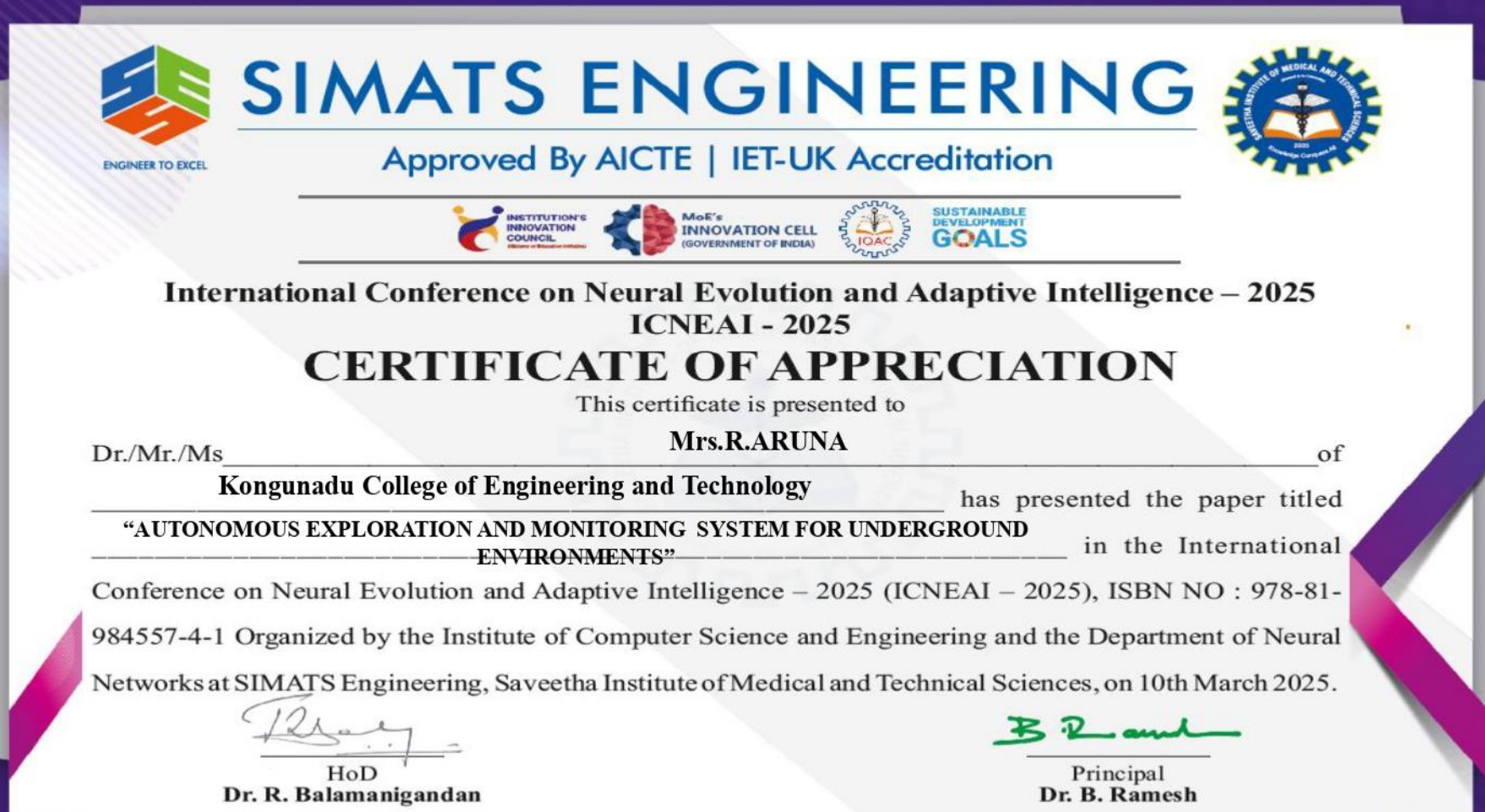

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